Manual

Basics of Hangul the Korean Writing System: A Guide for Beginners

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Abstract—This document serves as a comprehensive guide for beginners to learn the basics of Hangul, the Korean writing system. It covers the fundamental principles of Hangul, mainly its structure, and pronunciation. The guide is designed to provide learners with a solid foundation in reading and writing Hangul.

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1. INTRODUCTION

Contrary to a common misconception, Hangul is not a logographic writing system like Chinese characters. Instead, it is a featural alphabet consisting of 14 basic consonants and 10 basic vowels written in blocks to represent the syllables of the Korean language. It was created in 1443 by King Sejong the Great and his scholars with the purpose of promoting literacy among the common people, therefore it is characterized by its simplicity and ease of learning [1].

2. PRINCIPLES

It is crucial to understand the basic principles of Hangul before diving into its structure and pronunciation. The Korean writing system is based on a specific arrangement of consonants and vowels, that result in 2-dimensional blocks representing syllables, which are then written linearly to form words. This unusual arrangement of letters is probably the most challenging aspect of learning Hangul.

The consonants are designed to represent the shape of the speech organs used to produce them, while the vowels are designed to represent the philosophical concepts of the universe at the time of its creation [1]. While the visual appeal of combining letters into single syllabic blocks may vary from person to person, it cannot be denied that the clear visual distinction between consonants and vowels is a welcome sign for any reader.

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3. STRUCTURE

This will be the most crucial part of this guide, since it is the basis of every syllable in the Korean language. Every syllable is written as a single block built from up to three parts: an initial consonant c_i, a total vowel v_t, and a final optional consonant c_o.

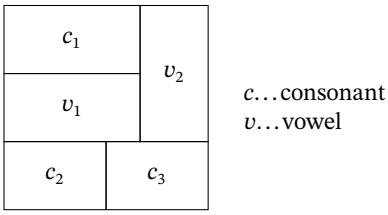


Figure 1. Basic structure of a Hangul syllable block

fig. 1 illustrates the basic structure of Hangul syllable blocks. The order of how to read the letters within a block is as follows:

$$\text{Syllable} = \underbrace{c_1}_{\text{initial consonant } c_i} + \underbrace{(v_1 + v_2)}_{\text{total vowel } v_t} + \underbrace{(c_2 + c_3)}_{\text{optional consonant } c_o} \tag{1}$$

c_o eq. (1) If the optional consonant c_o uses both c_2 and c_3, then they are spoken together as a single consonant sound by compressing the sound of the individual consonants.

4. CONSONANTS

As mentioned before the consonants are designed to represent the shape of the speech organs used to produce them [1]. Therefore the consonants are divided roughly into four groups (tongue, teeth, throat, and lips). Which have been further divided into subgroups based on the place of articulation, basically where the organs are placed when pronouncing the consonants. The following list shows the basic consonants and their corresponding English letter.

For convenience I will use the following notation to represent the consonants in this document (please use fig. 1 and eq. (1) as reference):

$$\begin{pmatrix} \text{hangul letter} \\ \text{pronunciation}_{\text{reference word}} \text{ as } c_1 \\ \text{pronunciation}_{\text{reference word}} \text{ as } c_2 \text{ or } c_3 \\ \text{special cases} \end{pmatrix} \tag{2}$$

The pronunciations given here are simplified for beginners, based on the official standard pronunciation rules [2].

Note that the pronunciations are written phonetically, as an English speaker would read them, rather than in the official Revised Romanization of Korean (e.g. ㅡ is written as “eww” instead of “eu”).

Onomatopoeia and interjection words will be marked with ” when used as reference words.

4.1. Tongue, velar

Back of tongue against the soft palate; ㄱ shows the tongue root.

$$\begin{pmatrix} \text{ㄱ} \\ g_{go} \\ g_{go} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅋ} \\ k_{kite} \\ g_{go} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㄲ} \\ gg_{\text{“Gulp!”}} \\ g_{go} \\ - \end{pmatrix} \tag{3}$$

4.2. Tongue, alveolar

Tongue tip touches the ridge behind the upper teeth; ㄴ shows the tongue.

$$\begin{pmatrix} \text{ㄴ} \\ \text{n}_{\textbf{no}} \\ \text{n}_{\textbf{no}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㄷ} \\ \text{d}_{\textbf{do}} \\ \text{d}_{\textbf{do}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅌ} \\ \text{t}_{\textbf{top}} \\ \text{d}_{\textbf{do}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㄸ} \\ \text{dd}_{\prime\prime\textbf{Ding!}\prime\prime} \\ - \\ - \end{pmatrix} \tag{4}$$

$$\begin{pmatrix} \text{ㄹ} \\ \text{r}_{\textbf{Ma}\text{ria}} \\ \text{l}_{\textbf{ball}} \\ \text{c}_{o,last}^{\text{prev block}} = \text{ㄹ} \;\Rightarrow\; \text{c}_1 = \text{l}_{\textbf{ball}} \end{pmatrix} \tag{5}$$

ㄹ eq. (5): "If the previous block ends in ㄹ, an initial ㄹ is also read as l (빨리 → ppalli)."

4.3. Lips, bilabial

Lips pressed together; ㅁ shows the mouth.

$$\begin{pmatrix} \text{ㅁ} \\ \text{m}_{\textbf{m}\text{oon}} \\ \text{m}_{\textbf{m}\text{oon}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅂ} \\ \text{b}_{\textbf{b}\text{oy}} \\ \text{b}_{\textbf{b}\text{oy}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅍ} \\ \text{p}_{\textbf{p}\text{ie}} \\ \text{b}_{\textbf{b}\text{oy}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅃ} \\ \text{bb}_{\prime\prime\textbf{B}\text{ang!}\prime\prime} \\ - \\ - \end{pmatrix} \tag{6}$$

4.4. Teeth, sibilant

Air hisses past the teeth; ㅅ shows the gap between teeth.

$$\begin{pmatrix} \text{ㅅ} \\ \text{s}_{\textbf{s}\text{un}} \\ \text{d}_{\textbf{d}\text{o}} \\ \left(\begin{smallmatrix} \text{c}_1 = \text{ㅅ} \\ \text{v}_t = \text{ㅣ} \end{smallmatrix}\right) \Rightarrow \text{c}_1 = \text{sh}_{\textbf{sh}\text{e}} \\ \left(\begin{smallmatrix} \text{c}_3 = \text{ㅅ} \\ \text{c}_1^{\text{next block}} = \text{ㅇ} \end{smallmatrix}\right) \Rightarrow \text{c}_3 = \text{s}_{\textbf{s}\text{un}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅈ} \\ \text{j}_{\textbf{j}\text{am}} \\ \text{d}_{\textbf{d}\text{o}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅊ} \\ \text{ch}_{\textbf{ch}\text{ild}} \\ \text{d}_{\textbf{d}\text{o}} \\ - \end{pmatrix} \tag{7}$$

$$\begin{pmatrix} \text{ㅆ} \\ \text{ss}_{\textbf{s}\text{ea}} \\ \text{d}_{\textbf{d}\text{o}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅉ} \\ \text{jj}_{\textbf{G}\text{u}\textbf{cci}} \\ - \\ - \end{pmatrix} \tag{8}$$

ㅅ eq. (7): "Before ㅣ, ㅅ is read as sh (시 → shi)."
ㅅ eq. (7): "If the next block starts with ㅇ, ㅅ is read as s (있어 → ids-uh)."

4.5. Throat, glottal

Sound from the throat; ㅇ shows the open throat

$$\begin{pmatrix} \text{ㅇ} \\ \varnothing \\ \text{ng}_{\textbf{s}\text{ing}} \\ - \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅎ} \\ \text{h}_{\textbf{h}\text{at}} \\ \varnothing \\ - \end{pmatrix} \tag{9}$$

∅ eq. (9) means silent

5. VOWELS

Traditionally the vowels were produced by placing the following 3 symbols in different relation to each other. ·, ㅡ, ㅣ, which each represented heaven, earth, human respectively [1]. While the philosophy behind it may be fascinating, for efficiency's sake it is easier to classify vowels in 3 groups. horizontal vowels v_1, vertical vowels v_2 and their combos $v_1 + v_2$.

For ease of use I will use the following convention:

$$\begin{pmatrix} \text{hangul letter} \\ \text{vowel}_{\text{reference word}} \end{pmatrix} \tag{10}$$

5.1. Horizontal Vowels v_1

They are vowels that have ㅡ (earth) as basis and fit into v_1.

$$\begin{pmatrix} \text{ㅡ} \\ \text{eww}_{\prime\prime\textbf{eww!}\prime\prime} \end{pmatrix} \tag{11}$$

$$\begin{pmatrix} \text{ㅗ} \\ \text{o}_{\textbf{go}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅛ} \\ \text{yo}_{\textbf{yo}\text{ga}} \end{pmatrix} \tag{12}$$

$$\begin{pmatrix} \text{ㅜ} \\ \text{oo}_{\textbf{moon}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅠ} \\ \text{you}_{\textbf{you}} \end{pmatrix} \tag{13}$$

5.2. Vertical Vowels v_2

They are vowels that have ㅣ (human) as basis and fit into v_2.

$$\begin{pmatrix} \text{ㅣ} \\ \text{i}_{\textbf{see}} \end{pmatrix} \tag{14}$$

$$\begin{pmatrix} \text{ㅏ} \\ \text{a}_{\text{spa}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅑ} \\ \text{ya}_{\textbf{ya}\text{rd}} \end{pmatrix} \tag{15}$$

$$\begin{pmatrix} \text{ㅓ} \\ \text{uh}_{\textbf{s}\text{un}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅕ} \\ \text{yuh}_{\textbf{you}\text{ng}} \end{pmatrix} \tag{16}$$

$$\begin{pmatrix} \text{ㅐ} \\ \text{ae}_{\textbf{c}\text{at}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅒ} \\ \text{yae}_{\textbf{yea}\text{h}} \end{pmatrix} \tag{17}$$

$$\begin{pmatrix} \text{ㅔ} \\ \text{e}_{\text{hey}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅖ} \\ \text{ye}_{\textbf{ya}\text{y}} \end{pmatrix} \tag{18}$$

5.3. Compound vowels $v_1 + v_2$

They are vowels that are a combination of a horizontal vowel v_1 and a vertical vowel v_2.

$$\begin{pmatrix} \text{ㅢ} \\ \text{ui}_{\prime\prime\textbf{eww!}\prime\prime + \textbf{ee}} \end{pmatrix} \tag{19}$$

$$\begin{pmatrix} \text{ㅚ} \\ \text{oe}_{\textbf{we}\text{t}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅘ} \\ \text{wa}_{\textbf{wa}\text{sabi}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅙ} \\ \text{wae}_{\textbf{wa}\text{gon}} \end{pmatrix} \tag{20}$$

$$\begin{pmatrix} \text{ㅟ} \\ \text{wi}_{\textbf{wee}\text{k}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅝ} \\ \text{wo}_{\textbf{wo}\text{nder}} \end{pmatrix} \bullet\!-\!\bullet \begin{pmatrix} \text{ㅞ} \\ \text{we}_{\textbf{we}\text{dding}} \end{pmatrix} \tag{21}$$

6. EXAMPLES

The following examples demonstrate the process of breaking down Hangul syllables into their constituent parts, allowing learners to understand the structure and pronunciation of Korean words.

$$\text{가} = \underbrace{\begin{pmatrix} \text{ㄱ} \\ \boxed{\text{c}_1} \\ \text{g} \end{pmatrix}}_{\text{c}_i} + \underbrace{\begin{pmatrix} \boxed{\text{v}_1} \end{pmatrix} + \begin{pmatrix} \text{ㅏ} \\ \boxed{\text{v}_2} \\ \text{a} \end{pmatrix}}_{\text{v}_t} + \underbrace{\begin{pmatrix} \boxed{\text{c}_2} \end{pmatrix} + \begin{pmatrix} \boxed{\text{c}_3} \end{pmatrix}}_{\text{c}_o} \tag{22}$$

$$= \begin{pmatrix} \text{ㄱ} \\ \boxed{\text{c}_i} \\ \text{g} \end{pmatrix} + \begin{pmatrix} \text{ㅏ} \\ \boxed{\text{v}_t} \\ \text{a} \end{pmatrix} + \begin{pmatrix} \boxed{\text{c}_o} \end{pmatrix} = \begin{pmatrix} \text{가} \\ \boxed{\text{syllable}} \\ \text{ga} \end{pmatrix} \tag{23}$$

$$\text{말} = \underbrace{\begin{pmatrix}\text{ㅁ}\\ \boxed{c_1}\\ \text{m}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\text{ㅏ}\\ \boxed{v_2}\\ \text{a}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\text{ㄹ}\\ \boxed{c_2}\\ \text{l}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{24}$$

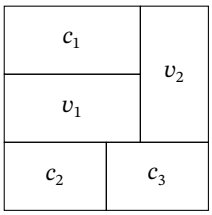
$$= \begin{pmatrix}\text{ㅁ}\\ \boxed{c_i}\\ \text{m}\end{pmatrix} + \begin{pmatrix}\text{ㅏ}\\ \boxed{v_t}\\ \text{a}\end{pmatrix} + \begin{pmatrix}\text{ㄹ}\\ \boxed{c_o}\\ \text{l}\end{pmatrix} = \begin{pmatrix}\text{말}\\ \boxed{\text{syllable}}\\ \text{mal}\end{pmatrix} \tag{25}$$

$$\text{왔} = \underbrace{\begin{pmatrix}\text{ㅇ}\\ \boxed{c_1}\\ \varnothing\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\text{ㅗ}\\ \boxed{v_1}\\ \text{o}\end{pmatrix} + \begin{pmatrix}\text{ㅏ}\\ \boxed{v_2}\\ \text{a}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\text{ㅅ}\\ \boxed{c_2}\\ \text{d}\end{pmatrix} + \begin{pmatrix}\text{ㅅ}\\ \boxed{c_3}\\ \text{d}\end{pmatrix}}_{c_o} \tag{26}$$

$$= \begin{pmatrix}\text{ㅇ}\\ \boxed{c_i}\\ \varnothing\end{pmatrix} + \begin{pmatrix}\text{ㅘ}\\ \boxed{v_t}\\ \text{wa}\end{pmatrix} + \begin{pmatrix}\text{ㅆ}\\ \boxed{c_o}\\ \text{d}\end{pmatrix} = \begin{pmatrix}\text{왔}\\ \boxed{\text{syllable}}\\ \text{wad}\end{pmatrix} \tag{27}$$

7. EXERCISES

Reminder fig. 1.



c…consonant
v…vowel

Reminder: Syllable Block

7.1. Consonant + vertical vowel

$$\text{나} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{28}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{na} \tag{29}$$

$$\text{개} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{30}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{gae} \tag{31}$$

7.2. Consonant + horizontal vowel

$$\text{도} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{32}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{do} \tag{33}$$

$$\text{그} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{34}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{g''eww''} \tag{35}$$

7.3. With an optional consonant

$$\text{밥} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{36}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{bab} \tag{37}$$

$$\text{물} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{38}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{mool} \tag{39}$$

7.4. Compound vowels

$$\text{왜} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{40}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{wae} \tag{41}$$

$$\text{뭐} = \underbrace{\begin{pmatrix}\boxed{c_1}\end{pmatrix}}_{c_i} + \underbrace{\begin{pmatrix}\boxed{v_1}\end{pmatrix} + \begin{pmatrix}\boxed{v_2}\end{pmatrix}}_{v_t} + \underbrace{\begin{pmatrix}\boxed{c_2}\end{pmatrix} + \begin{pmatrix}\boxed{c_3}\end{pmatrix}}_{c_o} \tag{42}$$

$$= \begin{pmatrix}\boxed{c_i}\end{pmatrix} + \begin{pmatrix}\boxed{v_t}\end{pmatrix} + \begin{pmatrix}\boxed{c_o}\end{pmatrix} = \begin{pmatrix}\boxed{\text{syllable}}\end{pmatrix} = \text{mwo} \tag{43}$$

7.5. Mixed

$$\text{닭} = \underbrace{\left(\boxed{c_1}\right)}_{c_i} + \underbrace{\left(\boxed{v_1}\right) + \left(\boxed{v_2}\right)}_{v_t} + \underbrace{\left(\boxed{c_2}\right) + \left(\boxed{c_3}\right)}_{c_o} \tag{44}$$

$$= \left(\boxed{c_i}\right) + \left(\boxed{v_t}\right) + \left(\boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right) = \text{dalg} \tag{45}$$

$$\text{읽} = \underbrace{\left(\boxed{c_1}\right)}_{c_i} + \underbrace{\left(\boxed{v_1}\right) + \left(\boxed{v_2}\right)}_{v_t} + \underbrace{\left(\boxed{c_2}\right) + \left(\boxed{c_3}\right)}_{c_o} \tag{46}$$

$$= \left(\boxed{c_i}\right) + \left(\boxed{v_t}\right) + \left(\boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right) = \text{ilg} \tag{47}$$

$$\text{없} = \underbrace{\left(\boxed{c_1}\right)}_{c_i} + \underbrace{\left(\boxed{v_1}\right) + \left(\boxed{v_2}\right)}_{v_t} + \underbrace{\left(\boxed{c_2}\right) + \left(\boxed{c_3}\right)}_{c_o} \tag{48}$$

$$= \left(\boxed{c_i}\right) + \left(\boxed{v_t}\right) + \left(\boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right) = \text{uhb} + (\text{s})_{c_1^{\text{next block}} = \circ} \tag{49}$$

$$\text{봤} = \underbrace{\left(\boxed{c_1}\right)}_{c_i} + \underbrace{\left(\boxed{v_1}\right) + \left(\boxed{v_2}\right)}_{v_t} + \underbrace{\left(\boxed{c_2}\right) + \left(\boxed{c_3}\right)}_{c_o} \tag{50}$$

$$= \left(\boxed{c_i}\right) + \left(\boxed{v_t}\right) + \left(\boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right) = \text{bwad} + (\text{s})_{c_1^{\text{next block}} = \circ} \tag{51}$$

$$\text{줬} = \underbrace{\left(\boxed{c_1}\right)}_{c_i} + \underbrace{\left(\boxed{v_1}\right) + \left(\boxed{v_2}\right)}_{v_t} + \underbrace{\left(\boxed{c_2}\right) + \left(\boxed{c_3}\right)}_{c_o} \tag{52}$$

$$= \left(\boxed{c_i}\right) + \left(\boxed{v_t}\right) + \left(\boxed{c_o}\right) = \left(\boxed{\text{syllable}}\right) = \text{jwod} + (\text{s})_{c_1^{\text{next block}} = \circ} \tag{53}$$

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